

1 PROGRAMME BASIS

Project	Underground CBRN-Hardened Blast-Resistant Protective Structure and Sentry Post
Location	Pune, Maharashtra
Design basis	Architectural Rev F · Structural Phase 2 Rev A + M1
Services packages	Drainage DR1 · HVAC HV1 · Schedule of Finishes FN1 · Structural CAD SC1
Programme revision	WM1 — the R0 master construction schedule, rebuilt
Data date / start	Monday 02 November 2026
Completion	Saturday 20 November 2027
Duration	326 working days (384 calendar days, 12.6 months)
Activities	279, of which 18 milestones, in a 3-level WBS
Logic links	404 finish-to-start, start-to-start, finish-to-finish and finish-to-finish links with lags
Critical activities	75 activities at total float 0
Calendar	Six-day week, Mon–Sat, 08:00–17:00 with a one-hour break

Six-day working week, Monday to Saturday; Sunday non-working. Five date-certain national holidays are non-working. Maharashtra festival holidays are NOT in the calendar because their dates move; the programme carries a stated contingency activity before handover to absorb them. Monsoon (June to September at Pune) is handled by productivity allowances inside the affected durations and by the sequencing rule that the deep excavation and the external tanking are kept out of the monsoon, not by making the calendar non-working.

2 THE SIX DECISIONS THAT SHAPE THIS PROGRAMME

- 1 Blast doors are ordered on day 1. The cast-in frames must be welded into the W6 and W7 cages before those walls are poured, so a 75-day frame lead time sits directly upstream of the wall concrete.
- 2 The pressure slab props stay for 14 days. IS 456 Table 11 requires 14 days for props to a slab spanning over 4.5 m; this one spans 5.000 m clear. Fourteen days of curing plus five of striking sit on the critical path and cannot be compressed.
- 3 The roof is cast before the walls are backfilled. Backfilling against a 3.2 m unpropped wall is what the 15.41 kPa/m lateral gradient forbids; the roof provides the prop.
- 4 The engineered cover is a six-layer structure, not a fill operation. It contains a 200 mm M30 burster slab that must be formed, reinforced, poured and cured inside the cover.
- 5 The sentry post starts only after the rock breaking finishes. No green concrete stands within 10 m of a hydraulic breaker, and the post carries 118 days of float, which makes it the natural place to absorb delay and redeploy labour.
- 6 Nothing is programmed against a fabricated festival-holiday date. The calendar carries only date-certain national holidays; a 10-day contingency activity before handover absorbs the movable ones.

5 WHAT THIS PROGRAMME ASSUMES, AND WHAT MUST BE CONFIRMED BEFORE IT CAN BE RELIED ON

WM-P1	Project start Monday 2 November 2026 is carried over from the supplied R0 master construction schedule. It is a planning date, not a contractual one.
WM-P2	Durations are built from the quantities derived in the BOQ and from output rates typical of Indian sites; they are ASSUMED and must be re-tested against the appointed contractor's resources.
WM-P3	The rockhead is ASSUMED in the master at (-)1.500 to (-)2.000. Rock excavation is programmed at the mean, 994 m3. At the shallow bound it rises to 1043 m3 and the excavation lengthens by about a day.
WM-P4	The design groundwater table (-)2.000 is ASSUMED. A1080 puts monsoon monitoring on the critical path deliberately: if the water is higher, the dewatering, the tanking and the flotation check all change.
WM-P5	No blasting is assumed. Rock is removed by hydraulic breaker because the sentry post founds on the same rock within 10 m.
WM-P6	Vendor lead times for the blast doors, filter trains, blast valves and the EMP enclosure are ASSUMED. They are the longest single risk in the programme and must be replaced by quoted times at tender.
WM-P7	Festival holidays are not in the calendar; A14098 carries ten working days of contingency in their place.
WM-P8	The berm, the access route and the concealment works have no site plan behind them, so their durations are indicative only.

3 MILESTONE REGISTER

REF	ACT	MILESTONE	DATE	FLOAT	STATUS
M-01	A1025	Site possession and commencement	07 Nov 26	0 d	CRITICAL
M-02	A1065	Mobilisation complete	26 Nov 26	8 d	
M-03	A1175	All long-lead protective items ordered	23 Feb 27	230 d	
M-04	A2090	Excavation complete, formation approved	06 Feb 27	0 d	CRITICAL
M-05	A3085	Mat foundation cast	15 Mar 27	213 d	
M-06	A3160	Perimeter and protective walls complete	05 May 27	170 d	
M-07	A4085	Pressure slab cast — key structural milestone	28 Jun 27	26 d	
M-08	A6055	Buried envelope watertight — tanking complete	05 Aug 27	91 d	
M-09	A7065	Overburden complete — structure buried	10 Sep 27	0 d	CRITICAL
M-10	A8125	Sentry post frame complete	20 Apr 27	182 d	
M-11	A8165	Sentry post brick masonry complete	28 Apr 27	175 d	
M-12	A8215	Sentry post complete	11 Jun 27	109 d	
M-13	A9060	CBRN plant installed	26 Aug 27	23 d	
M-14	A10065	Drainage installed	16 Sep 27	13 d	
M-15	A11060	Electrical installation complete	02 Sep 27	22 d	
M-16	A12070	Internal finishes complete	18 Sep 27	53 d	
M-17	A13050	External works and concealment complete	18 Oct 27	0 d	CRITICAL
M-18	A14105	PRACTICAL COMPLETION AND HANDOVER	20 Nov 27	0 d	CRITICAL

4 CRITICAL PATH — THE CONTROLLING CHAIN

- 1 PRE-CONSTRUCTION AND ENABLING WORKS — enters the path at A1010 on 02 Nov 26
- 2 SITE PREPARATION AND EARTHWORKS — enters the path at A2025 on 11 Dec 26
- 3 MAIN UNDERGROUND SHELTER — SUBSTRUCTURE — enters the path at A3010 on 06 Feb 27
- 4 MAIN UNDERGROUND SHELTER — PRESSURE SLAB — enters the path at A4010 on 05 May 27
- 6 WATERPROOFING — enters the path at A6030 on 23 Jul 27
- 7 BACKFILL, ENGINEERED COVER AND OVERBURDEN — enters the path at A7025 on 05 Aug 27
- 13 EXTERNAL WORKS, CONCEALMENT AND RESTORATION — enters the path at A13010 on 10 Sep 27
- 14 TESTING, COMMISSIONING AND HANDOVER — enters the path at A14080 on 18 Oct 27

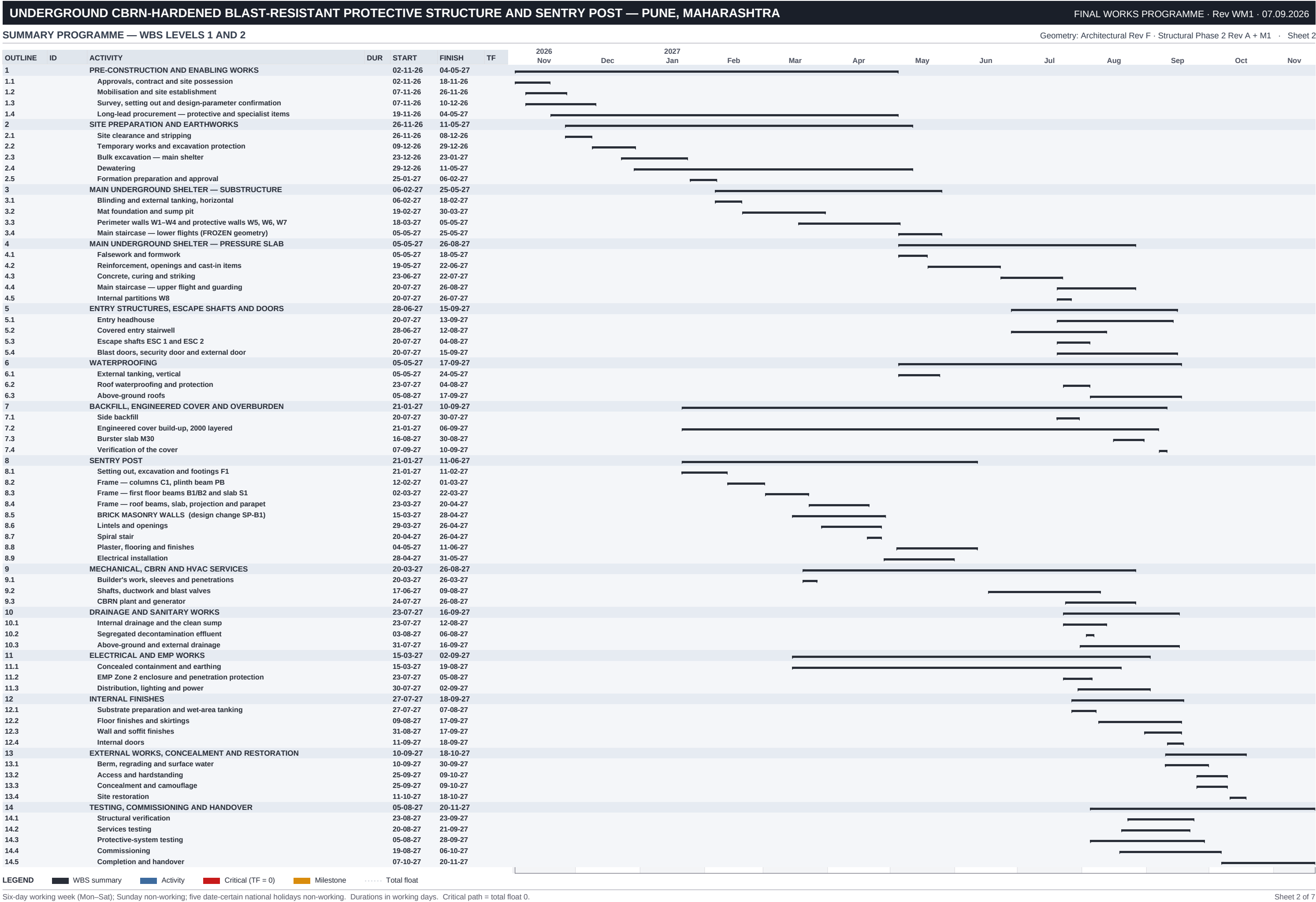
The controlling chain is: site possession and the confirmatory site investigation, which must close the ASSUMED groundwater level before the excavation support can be designed; rock excavation; formation approval; the external tanking; the mat; the perimeter and protective walls; the pressure slab with its 14-day prop period; the roof membrane; the six-layer engineered cover including the burster slab; the external works and concealment; and finally snagging, rectification and handover. Every one of the eighteen milestones is either on that chain or hangs from it.

6 PEAK RESOURCE REQUIREMENT AND FILE FORMAT

GROUP	RESOURCE	COMPOSITION	PEAK	DRIVEN BY
Labour	Bar bender gang	1 gang = 1 fitter + 3 helpers	4	Peak on the pressure slab: 20.5 t of T25 at 150 both faces both
Labour	Carpenter gang	1 gang = 1 carpenter + 2 helpers	4	Peak on the pressure slab falsework, 104 m2 of deck at 3.200 m
Labour	Concrete gang	1 gang = 1 mason + 6 helpers	2	Continuous pours: mat 85 m3, pressure slab 112 m3
Labour	Mason gang	1 gang = 1 mason + 2 helpers	3	SENTRY POST BRICK MASONRY; screeds; floor finishes; chambers
Labour	Plasterer gang	1 gang = 1 plasterer + 1 helper	2	Sentry post internal and external plaster
Labour	Painter gang	1 gang = 1 painter + 1 helper	2	Sealed washable coatings internally; sentry post painting
Labour	Waterproofing gang	specialist, 1 gang = 4 No.	4	Tanking is a specialist trade — 532 m2 of continuous membrane
Labour	Electrician gang	1 gang = 1 electrician + 1 helper	3	Conduits, cabling, earthing, EMP bonding
Labour	Plumber gang	1 gang = 1 plumber + 1 helper	2	Drainage, sumps, rising mains
Labour	HVAC fitter	1 gang = 1 fitter + 1 helper	2	Ductwork, blast valves, CBRN plant
Labour	Unskilled gang	general labour	20	Curing, handling, backfill, cleaning
Labour	Specialist subcontractor	as required	1	EMP enclosure, blast doors, CBRN commissioning, gas-tightness te
Plant	Excavator	0.9 m3 tracked	2	Bulk excavation, backfill, cover
Plant	Rock breaker	hydraulic, excavator mounted	2	994 m3 of basalt; NO BLASTING permitted near the sentry post
Plant	Tipper	10 t	4	Spoil haulage; peak during rock excavation
Plant	Dozer/grader	1 No.	1	Berm, regrading, access
Plant	Vibratory roller	1 No.	1	Backfill and cover compaction to 95 % MDD
Plant	Plate compactor	2 No.	2	Confined backfill in 250 layers
Plant	Concrete pump	boom pump	1	112 m3 continuous pour to the pressure slab
Plant	Transit mixer	6 m3	4	Peak on the pressure slab pour
Plant	Crane	20 t mobile	1	Blast doors, filter trains, EMP enclosure, spiral stair
Plant	Dewatering pumps	duty + standby	2	Continuous through the substructure; GWT (-)2.000 is ASSUMED
Plant	Scaffolding	tube and fitting	1	Slab falsework, masonry, plaster; IS 3696 / IS 4014
Plant	Welding set	2 No.	2	EMP straps and blast door frames welded to the reinforcement cag

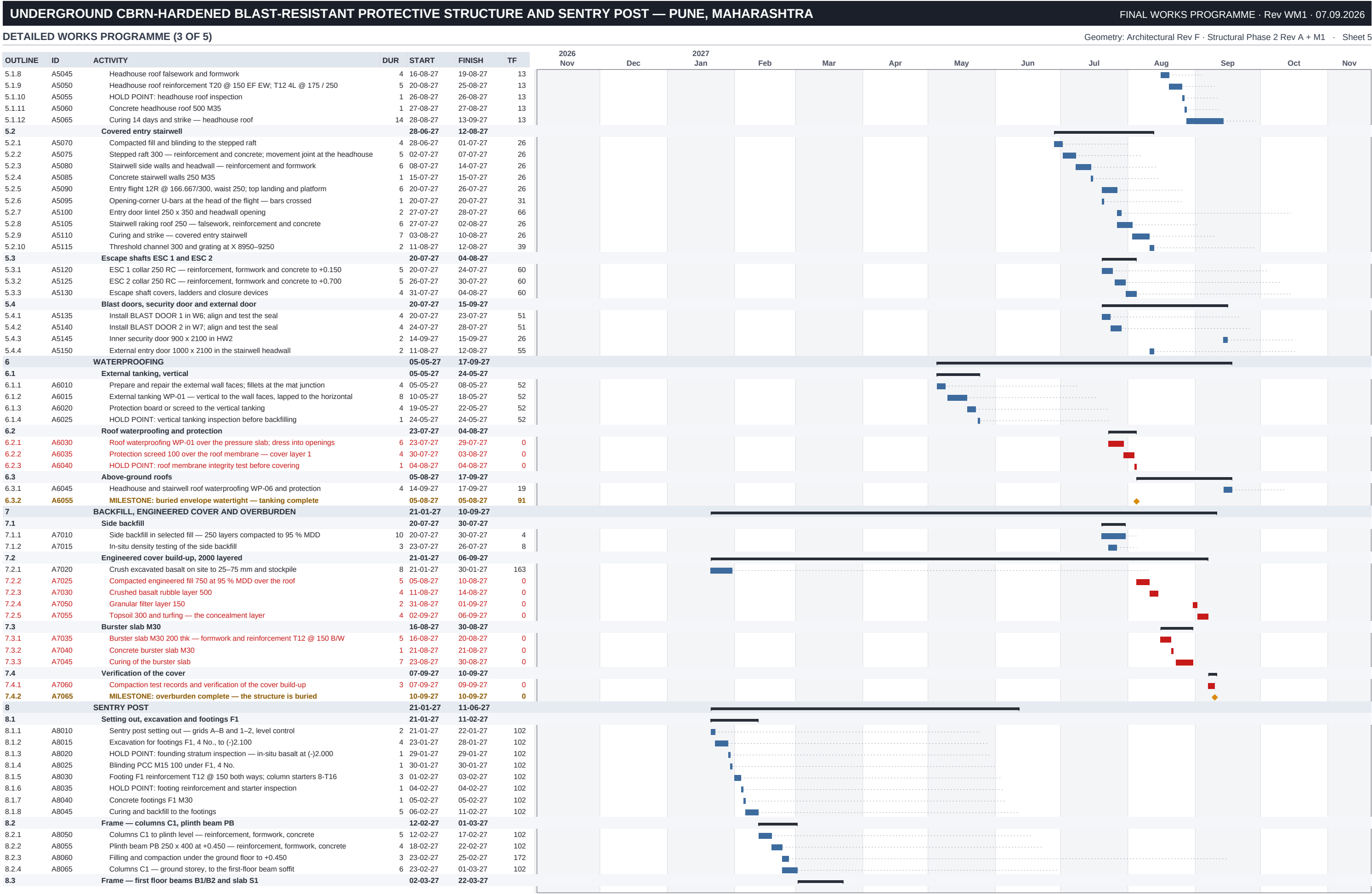
PROGRAMME FILE FORMAT

The editable master programme accompanying this PDF is Underground_Shelter_Final_Works_Programme.xml, written in MSPDI — Microsoft Project's own published XML schema. Microsoft Project opens it directly and File > Save As > Project (*.mpp) produces the binary .mpp. The native .mpp format is undocumented and can only be written by Microsoft Project itself, so it is not produced here rather than being imitated. The XML carries the full WBS, all 279 activities, all 404 logic links with their lags, the six-day calendar with its exceptions, 33 resources and 426 assignments; it was verified by reading it back with MPXJ and comparing every date against the critical-path calculation.



UNDERGROUND CBRN-HARDENED BLAST-RESISTANT PROTECTIVE STRUCTURE AND SENTRY POST — PUNE, MAHARASHTRA																FINAL WORKS PROGRAMME · Rev WM1 · 07.09.2026			
DETAILED WORKS PROGRAMME (1 OF 5)																Geometry: Architectural Rev F · Structural Phase 2 Rev A + M1 · Sheet 3			
OUTLINE	ID	ACTIVITY	DUR	START	FINISH	TF	2026	Dec	2027	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov
							Nov		Jan										
1		PRE-CONSTRUCTION AND ENABLING WORKS		02-11-26	04-05-27														
1.1		Approvals, contract and site possession		02-11-26	18-11-26														
1.1.1	A1010	Contract award, signature and site possession	5	02-11-26	06-11-26	0													
1.1.2	A1015	Statutory and Service approvals; security clearance of the workforce	10	02-11-26	12-11-26	19													
1.1.3	A1020	Contractor's programme, method statements and quality plan — approval	10	07-11-26	18-11-26	19													
1.1.4	A1025	MILESTONE: Site possession and commencement		07-11-26	07-11-26	0													
1.2		Mobilisation and site establishment		07-11-26	26-11-26														
1.2.1	A1030	Site office, stores, labour accommodation and first-aid post	8	07-11-26	16-11-26	8													
1.2.2	A1035	Site boundary, security fencing, gate and access control	6	07-11-26	13-11-26	13													
1.2.3	A1040	Temporary water supply, storage and distribution	5	10-11-26	14-11-26	17													
1.2.4	A1045	Temporary power supply and site distribution board	5	10-11-26	14-11-26	17													
1.2.5	A1050	Concrete supply arrangements and site testing laboratory	8	17-11-26	25-11-26	8													
1.2.6	A1055	Material stacking areas, reinforcement yard and bar-bending area	5	17-11-26	21-11-26	11													
1.2.7	A1060	Plant mobilisation — excavator, breakers, crane, pumps, compactors	5	14-11-26	19-11-26	13													
1.2.8	A1065	MILESTONE: Mobilisation complete		26-11-26	26-11-26	8													
1.3		Survey, setting out and design-parameter confirmation		07-11-26	10-12-26														
1.3.1	A1070	Topographic survey, benchmarks and establishment of the site grid	4	07-11-26	11-11-26	0													
1.3.2	A1075	Confirmatory site investigation — boreholes, rockhead, red-bole seams	12	12-11-26	25-11-26	0													
1.3.3	A1080	Monsoon groundwater monitoring — confirm the design GWT (-)2.000	20	12-11-26	04-12-26	0													
1.3.4	A1085	Percolation test for the soak pits — IS 2470 (Part 2) Cl. 4	5	26-11-26	01-12-26	3													
1.3.5	A1090	Issue confirmed geotechnical parameters to the designer	5	05-12-26	10-12-26	0													
1.3.6	A1095	Setting out — shelter box, headhouse and covered stairwell	3	12-11-26	14-11-26	13													
1.3.7	A1100	Setting out — sentry post, not less than 10 m clear of the excavation	2	16-11-26	17-11-26	13													
1.3.8	A1105	Setting-out check and independent verification	2	18-11-26	19-11-26	13													
1.4		Long-lead procurement — protective and specialist items		19-11-26	04-05-27														
1.4.1	A1110	Blast doors 1 and 2 — specification, enquiry and technical evaluation	15	19-11-26	05-12-26	19													
1.4.2	A1115	Blast doors — order placement	5	07-12-26	11-12-26	19													
1.4.3	A1120	Blast door cast-in FRAMES — manufacture and delivery	75	12-12-26	11-03-27	19													
1.4.4	A1122	Blast door LEAVES — manufacture, factory proof test and shipment	120	12-12-26	04-05-27	116													
1.4.5	A1130	CBRN filter trains AHU-1/AHU-2 — enquiry and order	20	19-11-26	11-12-26	115													
1.4.6	A1135	CBRN filter trains — manufacture and delivery	100	12-12-26	09-04-27	115													
1.4.7	A1140	Blast valves, 5 No. — enquiry and order	20	19-11-26	11-12-26	131													
1.4.8	A1145	Blast valves and sleeves — manufacture and delivery	100	12-12-26	09-04-27	131													
1.4.9	A1150	EMP Zone 2 shielded enclosure — specialist enquiry and order	25	19-11-26	17-12-26	148													
1.4.10	A1155	EMP Zone 2 enclosure — manufacture and delivery	80	18-12-26	23-03-27	148													
1.4.11	A1160	Generator 15 kVA — enquiry, order and delivery	60	19-11-26	29-01-27	195													
1.4.12	A1165	Armoured vision panels, 8 No. — enquiry, order and delivery	80	19-11-26	22-02-27	173													
1.4.13	A1170	Submersible pumps and drainage plant — order and delivery	45	19-11-26	11-01-27	214													
1.4.14	A1172	Reinforcement Fe500D — order, mill certificates and delivery	30	19-11-26	23-12-26	47													
1.4.15	A1174	Waterproofing system — approval of the proprietary system and order	30	19-11-26	23-12-26	39													
1.4.16	A1176	Modular bricks to IS 1077 — source approval, testing and order	20	19-11-26	11-12-26	198													
1.4.17	A1175	MILESTONE: All long-lead protective items ordered		23-02-27	23-02-27	230													
2		SITE PREPARATION AND EARTHWORKS		26-11-26	11-05-27														
2.1		Site clearance and stripping		26-11-26	08-12-26														
2.1.1	A2010	Site clearance, grubbing and removal of plants and shrubs	4	26-11-26	30-11-26	8													
2.1.2	A2015	Strip topsoil 150 and stockpile for the concealment layer	3	01-12-26	03-12-26	8													
2.1.3	A2020	Site levelling, temporary haul road and hardstanding	4	04-12-26	08-12-26	8													
2.2		Temporary works and excavation protection		09-12-26	29-12-26														
2.2.1	A2025	Design and approval of excavation support and slope treatment	10	11-12-26	22-12-26	0													
2.2.2	A2030	Surface water cut-off drains and interceptor around the excavation	4	09-12-26	12-12-26	8													
2.2.3	A2035	Excavation edge protection, barriers, signage and access ladders	3	26-12-26	29-12-26	30													
2.3		Bulk excavation — main shelter		23-12-26	23-01-27														
2.3.1	A2040	Excavation in soil and weathered overburden to rockhead	4	23-12-26	28-12-26	0													
2.3.2	A2045	Rock excavation by hydraulic breaker, (-)1.750 to (-)4.000	9	29-12-26	07-01-27	0													
2.3.3	A2050	Rock excavation by hydraulic breaker, (-)4.000 to (-)6.800	11	08-01-27	20-01-27	0													
2.3.4	A2055	Excavation for the covered entry stairwell approach	5	14-01-27	19-01-27	160													
2.3.5	A2060	Excavation in rock for the sump pit, to (-)8.100	3	21-01-27	23-01-27	0													
2.4		Dewatering		29-12-26	11-05-27														
2.4.1	A2065	Install and commission the construction dewatering system	3	29-12-26	31-12-26	73													
2.4.2	A2070	Dewatering — continuous through the substructure works	110	01-01-27	11-05-27	73													
2.5		Formation preparation and approval		25-01-27	06-02-27														
2.5.1	A2075	Trim, dress and clean the formation; remove loose rock	4	25-01-27	29-01-27	0													
2.5.2	A2080	Inspect for red-bole and vesicular seams; over-excavate and replace with M15	4	30-01-27	03-02-27	0													
2.5.3	A2085	HOLD POINT: formation inspection and approval by the Engineer	2	04-02-27	05-02-27	0													
2.5.4	A2090	MILESTONE: excavation complete, formation approved		06-02-27	06-02-27	0													
3		MAIN UNDERGROUND SHELTER — SUBSTRUCTURE		06-02-27	25-05-27														
3.1		Blinding and external tanking, horizontal		06-02-27	18-02-27														
3.1.1	A3010	Blinding PCC M15 100 thk under the mat	3	06-02-27	09-02-27	0													
3.1.2	A3015	Blinding PCC M15 to the sump pit	1	06-02-27	06-02-27	2													
3.1.3	A3020	External tanking WP-01 — horizontal, on the blinding	5	10-02-27	15-02-27	0													
3.1.4	A3025	Protection layer to the horizontal tanking	2	16-02-27	17-02-27	0													
LEGEND							WBS summary	Activity	Critical (TF = 0)	Milestone	Total float								

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DETAILED WORKS PROGRAMME (2 OF 5)										Geometry: Architectural Rev F · Structural Phase 2 Rev A + M1 · Sheet 4									
OUTLINE	ID	ACTIVITY	DUR	START	FINISH	TF	2026 Nov	Dec	2027 Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov
3.1.5	A3030	HOLD POINT: tanking integrity inspection before covering	1	18-02-27	18-02-27	0													
3.2		Mat foundation and sump pit		19-02-27	30-03-27														
3.2.1	A3035	Sump pit SU-01 — formwork and reinforcement below the mat	4	19-02-27	23-02-27	13													
3.2.2	A3040	Mat reinforcement — bottom curtain T16 @ 150 each way	4	19-02-27	23-02-27	0													
3.2.3	A3045	Wall starter bars, 900 leg into the mat; kickers set out	3	24-02-27	26-02-27	0													
3.2.4	A3050	Mat links T12 @ 250 x 250 grid, chairs and spacers	3	27-02-27	02-03-27	0													
3.2.5	A3055	Mat reinforcement — top curtain T16 @ 150 each way	4	03-03-27	06-03-27	0													
3.2.6	A3060	Mat edge formwork and U-bars; waterstops at the kicker	3	08-03-27	10-03-27	0													
3.2.7	A3065	Cast-in items — drainage BW-01, sleeves, earth pits, EMP straps	3	08-03-27	10-03-27	0													
3.2.8	A3070	HOLD POINT: mat reinforcement, cover and cast-in inspection	1	11-03-27	11-03-27	0													
3.2.9	A3075	Concrete mat 600 M35 and sump pit — single continuous pour	2	12-03-27	13-03-27	0													
3.2.10	A3080	Curing of the mat — 14 days continuous	14	15-03-27	30-03-27	1													
3.2.11	A3085	MILESTONE: mat foundation cast		15-03-27	15-03-27	213													
3.3		Perimeter walls W1–W4 and protective walls W5, W6, W7		18-03-27	05-05-27														
3.3.1	A3090	Wall reinforcement W1–W4 — both curtains T16 @ 150 each face, each way	12	18-03-27	31-03-27	0													
3.3.2	A3095	Wall reinforcement W5, W6, W7 incl. blast door jambs and header	6	25-03-27	31-03-27	0													
3.3.3	A3100	Haunch reinforcement 500 x 500, diagonal T20 @ 150, wall/mat junction	3	20-03-27	23-03-27	7													
3.3.4	A3105	Set and align the BLAST DOOR FRAMES in W6 and W7; weld to the cage	4	01-04-27	05-04-27	2													
3.3.5	A3110	Concealed conduits, sleeves and penetrations cast into the walls	5	01-04-27	06-04-27	1													
3.3.6	A3115	Wall formwork — lift 1, (-)6.100 to (-)4.500	6	01-04-27	07-04-27	0													
3.3.7	A3120	HOLD POINT: wall reinforcement, frames and cover inspection — lift 1	1	08-04-27	08-04-27	0													
3.3.8	A3125	Concrete walls — lift 1, M35	2	09-04-27	10-04-27	0													
3.3.9	A3130	Curing lift 1; strike vertical formwork after 24 h minimum	7	12-04-27	19-04-27	0													
3.3.10	A3135	Wall formwork — lift 2, (-)4.500 to (-)2.900	6	15-04-27	21-04-27	0													
3.3.11	A3140	HOLD POINT: wall reinforcement and cover inspection — lift 2	1	22-04-27	22-04-27	0													
3.3.12	A3145	Concrete walls — lift 2, M35	2	23-04-27	24-04-27	0													
3.3.13	A3150	Curing lift 2 and strike wall formwork	7	26-04-27	04-05-27	0													
3.3.14	A3155	Construction joint preparation — two waterstops and welded EMP strap	3	12-04-27	14-04-27	6													
3.3.15	A3160	MILESTONE: perimeter and protective walls complete		05-05-27	05-05-27	170													
3.4		Main staircase — lower flights (FROZEN geometry)		05-05-27	25-05-27														
3.4.1	A3165	Main staircase — formwork to flights 1 and 2 and landings L1, L2	5	05-05-27	10-05-27	113													
3.4.2	A3170	Main staircase — reinforcement, flights T12 @ 150, landings T12 @ 125	4	11-05-27	14-05-27	113													
3.4.3	A3175	Main staircase — concrete flights 1 and 2 and landings, M35	2	15-05-27	17-05-27	113													
3.4.4	A3180	Main staircase — curing and strike, lower flights	7	18-05-27	25-05-27	113													
4		MAIN UNDERGROUND SHELTER — PRESSURE SLAB		05-05-27	26-08-27														
4.1		Falsework and formwork		05-05-27	18-05-27														
4.1.1	A4010	Falsework and soffit deck to the pressure slab, 3.200 m height	8	05-05-27	13-05-27	0													
4.1.2	A4015	Edge formwork, stair void and escape shaft box-outs	4	14-05-27	18-05-27	0													
4.2		Reinforcement, openings and cast-in items		19-05-27	22-06-27														
4.2.1	A4020	Pressure slab — bottom curtain T25 @ 150 each way	6	19-05-27	25-05-27	0													
4.2.2	A4025	Opening trimmers — stair void 6-T25 top and bottom; re-entrant corners 4-T25 at	4	26-05-27	29-05-27	0													
4.2.3	A4030	Escape shaft collar trimmers 5-T25 each side/face/direction; hoops and radials	4	31-05-27	03-06-27	0													
4.2.4	A4035	Local thickenings 900 to 1200 at the shafts and the void free edge	3	01-06-27	03-06-27	16													
4.2.5	A4040	Band beneath headhouse wall HW3 — 4-T25 extra top and bottom in a 1200 band	2	31-05-27	01-06-27	18													
4.2.6	A4045	Links T12 4-leg @ 250 in the end 1500; 2-leg @ 300 mid	5	04-06-27	09-06-27	0													
4.2.7	A4050	Pressure slab — top curtain T25 @ 150 each way	6	10-06-27	16-06-27	0													
4.2.8	A4055	Wall/roof haunches 500 x 500, diagonal T20 @ 150	3	12-06-27	15-06-27	6													
4.2.9	A4060	Cast-in — blast valve sleeves, duct and pipe penetrations, EMP straps	5	17-06-27	22-06-27	0													
4.2.10	A4065	Escape shaft collar starter bars	2	17-06-27	18-06-27	3													
4.2.11	A4067	Concealed electrical conduits and boxes cast into the slab	4	07-06-27	10-06-27	10													
4.3		Concrete, curing and striking		23-06-27	22-07-27														
4.3.1	A4070	HOLD POINT: pressure slab pre-pour inspection — reinforcement, cover, cast-ins	2	23-06-27	24-06-27	0													
4.3.2	A4075	Concrete pressure slab 900 M35 — single continuous pour, 112 m3	2	25-06-27	26-06-27	0													
4.3.3	A4080	Curing of the pressure slab — 14 days continuous	14	28-06-27	13-07-27	0													
4.3.4	A4085	MILESTONE: pressure slab cast — the key structural milestone		28-06-27	28-06-27	26													
4.3.5	A4090	Strike soffit formwork and props — props to remain 14 days	5	14-07-27	19-07-27	0													
4.3.6	A4095	Make good, hack and prepare the soffit	3	20-07-27	22-07-27	0													
4.4		Main staircase — upper flight and guarding		20-07-27	26-08-27														
4.4.1	A4110	Main staircase — flight 3 to the top of the pressure slab (-)2.000	6	20-07-27	26-07-27	66													
4.4.2	A4115	Main staircase — handrails and guarding 1100 high	4	23-08-27	26-08-27	43													
4.5		Internal partitions W8		20-07-27	26-07-27														
4.5.1	A4100	Internal partitions W8, 4 No., 110 thk with A252 mesh both faces	6	20-07-27	26-07-27	7													
5		ENTRY STRUCTURES, ESCAPE SHAFTS AND DOORS		28-06-27	15-09-27														
5.1		Entry headhouse		20-07-27	13-09-27														
5.1.1	A5010	Headhouse — wall starters, kicker and setting out on the pressure slab	3	20-07-27	22-07-27	13													
5.1.2	A5015	Headhouse wall reinforcement HW1–HW4, T16 @ 150 EF EW, T12 4L @ 250	6	23-07-27	29-07-27	13													
5.1.3	A5020	Inner security door frame in HW2 — cast in and welded to the cage	2	30-07-27	31-07-27	13													
5.1.4	A5025	Headhouse wall formwork	5	02-08-27	06-08-27	13													
5.1.5	A5030	HOLD POINT: headhouse wall inspection	1	07-08-27	07-08-27	13													
5.1.6	A5035	Concrete headhouse walls 400 M35	1	09-08-27	09-08-27	13													
5.1.7	A5040	Curing and strike — headhouse walls	5	10-08-27	14-08-27	13													
LEGEND							■ WBS summary ■ Activity ■ Critical (TF = 0) ■ Milestone Total float												



UNDERGROUND CBRN-HARDENED BLAST-RESISTANT PROTECTIVE STRUCTURE AND SENTRY POST — PUNE, MAHARASHTRA										FINAL WORKS PROGRAMME · Rev WM1 · 07.09.2026									
DETAILED WORKS PROGRAMME (4 OF 5)										Geometry: Architectural Rev F · Structural Phase 2 Rev A + M1 · Sheet 6									
OUTLINE	ID	ACTIVITY	DUR	START	FINISH	TF	2026 Nov	Dec	2027 Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov
8.3.1	A8070	First floor beams B1/B2 and slab S1 150 — falsework and formwork	5	02-03-27	06-03-27	102													
8.3.2	A8075	First floor beam and slab reinforcement, incl. corner torsion steel	4	08-03-27	11-03-27	102													
8.3.3	A8080	HOLD POINT: first floor reinforcement inspection — IS 13920 detailing	1	12-03-27	12-03-27	102													
8.3.4	A8085	Concrete first floor beams and slab M30	1	13-03-27	13-03-27	102													
8.3.5	A8090	Curing 7 days; strike sides, props to remain	7	15-03-27	22-03-27	102													
8.4		Frame — roof beams, slab, projection and parapet	23-03-27	20-04-27															
8.4.1	A8095	Columns C1 — first storey, to the roof beam soffit	6	23-03-27	29-03-27	102													
8.4.2	A8100	Roof beams B1/B2 and slab S1 with the 300 projection — formwork and reinforce	6	30-03-27	05-04-27	102													
8.4.3	A8105	HOLD POINT: sentry post roof reinforcement inspection	1	06-04-27	06-04-27	102													
8.4.4	A8110	Concrete sentry post roof M30	1	07-04-27	07-04-27	102													
8.4.5	A8115	Parapet 300 high over the roof projection	3	12-04-27	14-04-27	186													
8.4.6	A8120	Curing and strike — sentry post roof	10	08-04-27	19-04-27	102													
8.4.7	A8125	MILESTONE: sentry post frame complete	20-04-27	20-04-27	182														
8.5		BRICK MASONRY WALLS (design change SP-B1)	15-03-27	28-04-27															
8.5.1	A8130	Brick delivery, stacking, sampling and testing to IS 3495	4	15-03-27	18-03-27	121													
8.5.2	A8135	BRICK MASONRY WALLS — ground storey, 190 thk in CM 1:6	5	23-03-27	27-03-27	118													
8.5.3	A8145	HOLD POINT: masonry line, level, plumb and joint thickness — ground storey	1	31-03-27	31-03-27	118													
8.5.4	A8150	BRICK MASONRY WALLS — first storey, 190 thk in CM 1:6	4	20-04-27	23-04-27	102													
8.5.5	A8160	HOLD POINT: masonry inspection — first storey	1	27-04-27	27-04-27	102													
8.5.6	A8165	MILESTONE: sentry post brick masonry complete	28-04-27	28-04-27	175														
8.6		Lintels and openings	29-03-27	26-04-27															
8.6.1	A8140	RC lintels over the D1 and W1 openings — ground storey	2	29-03-27	30-03-27	118													
8.6.2	A8155	RC lintels over the vision panel and door openings — first storey	2	24-04-27	26-04-27	102													
8.7		Spiral stair	20-04-27	26-04-27															
8.7.1	A8170	External spiral stair 1000 R, 250 dia central pole — fabrication and erection	6	20-04-27	26-04-27	147													
8.8		Plaster, flooring and finishes	04-05-27	11-06-27															
8.8.1	A8180	Sentry post internal plaster 12 mm, sand faced, CM 1:4	6	04-05-27	10-05-27	102													
8.8.2	A8185	Sentry post external plaster 15 mm, two coat, with scaffolding	6	11-05-27	17-05-27	102													
8.8.3	A8190	Sentry post flooring — ground and first floor	4	11-05-27	14-05-27	110													
8.8.4	A8195	Door D1 (2 No.), window W1 and armoured vision panel installation	6	18-05-27	24-05-27	102													
8.8.5	A8205	Sentry post painting — internal and external	6	01-06-27	07-06-27	102													
8.8.6	A8210	Sentry post testing and snagging	3	08-06-27	10-06-27	109													
8.8.7	A8215	MILESTONE: sentry post complete	11-06-27	11-06-27	109														
8.9		Electrical installation	28-04-27	31-05-27															
8.9.1	A8175	Sentry post electrical — conduits and boxes chased into the masonry	4	28-04-27	03-05-27	102													
8.9.2	A8200	Sentry post electrical — wiring, distribution board, fittings, earthing	6	25-05-27	31-05-27	102													
9		MECHANICAL, CBRN AND HVAC SERVICES	20-03-27	26-08-27															
9.1		Builder's work, sleeves and penetrations	20-03-27	26-03-27															
9.1.1	A9010	Builder's work — sleeves, ducts and penetrations coordinated with the RC	6	20-03-27	26-03-27	10													
9.2		Shafts, ductwork and blast valves	17-06-27	09-08-27															
9.2.1	A9015	Install blast valve sleeves in the walls and roof	4	17-06-27	21-06-27	74													
9.2.2	A9020	Fresh-air shaft SH-1 600 x 600 and gooseneck head at +1.500	5	20-07-27	24-07-27	45													
9.2.3	A9025	Generator air shaft SH-2 600 x 600	4	20-07-27	23-07-27	46													
9.2.4	A9030	Ductwork installation, bays 1–6 — every run inspectable along its length	10	23-07-27	03-08-27	17													
9.2.5	A9035	Install blast valves BV-1 to BV-5	5	04-08-27	09-08-27	37													
9.3		CBRN plant and generator	24-07-27	26-08-27															
9.3.1	A9040	Install CBRN filter trains AHU-1 and AHU-2 in bay 5	8	04-08-27	12-08-27	17													
9.3.2	A9045	Install CO2 scrubber, oxygen store and dehumidifier	5	13-08-27	18-08-27	17													
9.3.3	A9050	Install generator GEN-1 15 kVA in bay 8 on a bundled plinth	5	24-07-27	29-07-27	46													
9.3.4	A9055	HVAC terminal devices, dampers and controls	6	19-08-27	25-08-27	17													
9.3.5	A9060	MILESTONE: CBRN plant installed	26-08-27	26-08-27	23														
10		DRAINAGE AND SANITARY WORKS	23-07-27	16-09-27															
10.1		Internal drainage and the clean sump	23-07-27	12-08-27															
10.1.1	A10010	Internal gullies GY-01 to GY-09; floor falls set out	4	23-07-27	27-07-27	12													
10.1.2	A10015	Waste and effluent drains PD-01 and PD-03 — laid and tested	5	28-07-27	02-08-27	12													
10.1.3	A10020	Clean sump SU-01 — fit out, pumps PU-01/PU-02 and hand pump PU-03	5	03-08-27	07-08-27	42													
10.1.4	A10025	Rising main PD-05 — isolation valve, gas-tight NRV, blast check valve and deep-s	4	09-08-27	12-08-27	42													
10.2		Segregated decontamination effluent	03-08-27	06-08-27															
10.2.1	A10030	Decontamination effluent segregation — CP-01 and tank TK-01 in bay 8	4	03-08-27	06-08-27	47													
10.3		Above-ground and external drainage	31-07-27	16-09-27															
10.3.1	A10035	Stairwell sump SU-02 and pumps PU-04/PU-05	4	11-08-27	14-08-27	40													
10.3.2	A10040	Headhouse gully GY-11 and trapped connection	2	14-09-27	15-09-27	13													
10.3.3	A10045	Threshold channel CH-10, catchpit CP-10 and connections	3	13-08-27	16-08-27	39													
10.3.4	A10050	Septic tank ST-01 — excavation, construction and testing	8	31-07-27	09-08-27	33													
10.3.5	A10055	Soak pits SK-01 and SK-02 — constructed after the percolation test	6	10-08-27	16-08-27	33													
10.3.6	A10060	External drainage connections and chambers	6	17-08-27	23-08-27	33													
10.3.7	A10065	MILESTONE: drainage installed	16-09-27	16-09-27	13														
11		ELECTRICAL AND EMP WORKS	15-03-27	02-09-27															
11.1		Concealed containment and earthing	15-03-27	19-08-27															
11.1.1	A11015	Earth pits, earth electrodes and the buried earth ring	5	15-03-27	19-03-27	157													
11.1.2	A11020	EMP straps and bonding at every construction joint	8	09-04-27	17-04-27	138													
LEGEND							■ WBS summary ■ Activity ■ Critical (TF = 0) ■ Milestone Total float												

UNDERGROUND CBRN-HARDENED BLAST-RESISTANT PROTECTIVE STRUCTURE AND SENTRY POST — PUNE, MAHARASHTRA																FINAL WORKS PROGRAMME · Rev WM1 · 07.09.2026				
DETAILED WORKS PROGRAMME (5 OF 5)																Geometry: Architectural Rev F · Structural Phase 2 Rev A + M1 · Sheet 7				
OUTLINE	ID	ACTIVITY	DUR	START	FINISH	TF	2026 Nov	Dec	2027 Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	
11.1.3	A11050	Earthing conductors, bonding and connection to the electrode system	6	13-08-27	19-08-27	33														
11.2		EMP Zone 2 enclosure and penetration protection		23-07-27	05-08-27															
11.2.1	A11025	Main cable route, service entry plate and EMP penetration protection	6	23-07-27	29-07-27	12														
11.2.2	A11030	EMP Zone 2 shielded enclosure erection in bay 3 — SPECIALIST	12	23-07-27	05-08-27	45														
11.3		Distribution, lighting and power		30-07-27	02-09-27															
11.3.1	A11035	Distribution board, sub-mains and final circuit wiring	12	30-07-27	12-08-27	12														
11.3.2	A11040	Lighting, small power and emergency lighting installation	10	13-08-27	24-08-27	12														
11.3.3	A11045	Generator connection, changeover panel and protection	5	13-08-27	18-08-27	34														
11.3.4	A11055	Power to pumps, fans, filter trains and controls	6	26-08-27	01-09-27	17														
11.3.5	A11060	MILESTONE: electrical installation complete		02-09-27	02-09-27	22														
12		INTERNAL FINISHES		27-07-27	18-09-27															
12.1		Substrate preparation and wet-area tanking		27-07-27	07-08-27															
12.1.1	A12010	Substrate preparation — hack, make good, prepare soffits and walls	6	27-07-27	02-08-27	7														
12.1.2	A12015	Internal wet-area tanking WP-04 — bays 2, 5 and 6	5	03-08-27	07-08-27	7														
12.2		Floor finishes and skirtings		09-08-27	17-09-27															
12.2.1	A12020	Floor screeds laid to falls, bays 1–8	8	09-08-27	17-08-27	7														
12.2.2	A12025	Floor finish F-01 — sealed power-floated, bays 1, 3 and 4	6	18-08-27	24-08-27	7														
12.2.3	A12030	Floor finish F-02 — seamless coved wet-area, bays 2, 5 and 6	5	18-08-27	23-08-27	8														
12.2.4	A12035	Floor finish F-03 — non-slip, stair shaft, treads and landings	4	18-08-27	21-08-27	43														
12.2.5	A12040	Floor finish F-04 — oil-resistant with a banded plinth, bay 8	3	18-08-27	20-08-27	77														
12.2.6	A12045	Floor finish F-05 headhouse and F-06 covered entry stairwell	4	14-09-27	17-09-27	53														
12.2.7	A12050	Coved skirtings S-01 / S-02, integral with the floor	5	25-08-27	30-08-27	7														
12.3		Wall and soffit finishes		31-08-27	17-09-27															
12.3.1	A12055	Wall finishes W-01 / W-02 / W-03 — sealed washable coatings	10	31-08-27	10-09-27	7														
12.3.2	A12060	Soffit finishes C-01 / C-02 — exposed RC, sealed	6	11-09-27	17-09-27	7														
12.4		Internal doors		11-09-27	18-09-27															
12.4.1	A12065	Internal doors D-04 in the W8 partitions; gas-tight door D-05 in W5	4	11-09-27	15-09-27	55														
12.4.2	A12070	MILESTONE: internal finishes complete		18-09-27	18-09-27	53														
13		EXTERNAL WORKS, CONCEALMENT AND RESTORATION		10-09-27	18-10-27															
13.1		Berm, regrading and surface water		10-09-27	30-09-27															
13.1.1	A13010	Berm forming and grading 1.5:1 to +0.900 against the headhouse and stairwell	8	10-09-27	18-09-27	0														
13.1.2	A13015	Site regrading — crown and falls 1:50 away from the structure	5	20-09-27	24-09-27	0														
13.1.3	A13020	Surface water drainage, cut-off drains and outfalls	5	25-09-27	30-09-27	7														
13.2		Access and hardstanding		25-09-27	09-10-27															
13.2.1	A13025	Access route, hardstanding and turning area	8	25-09-27	05-10-27	0														
13.2.2	A13030	Sentry post surround, apron and drainage	4	06-10-27	09-10-27	0														
13.3		Concealment and camouflage		25-09-27	09-10-27															
13.3.1	A13035	Turfing and planting over the cover — the concealment layer	6	25-09-27	01-10-27	0														
13.3.2	A13040	Camouflage and concealment — spoil dressing, track discipline, screening	6	04-10-27	09-10-27	0														
13.4		Site restoration		11-10-27	18-10-27															
13.4.1	A13045	Removal of temporary works, site restoration and clearing	6	11-10-27	16-10-27	0														
13.4.2	A13050	MILESTONE: external works and concealment complete		18-10-27	18-10-27	0														
14		TESTING, COMMISSIONING AND HANDOVER		05-08-27	20-11-27															
14.1		Structural verification		23-08-27	23-09-27															
14.1.1	A14010	Concrete cube results at 28 days, all pours — review and acceptance	5	23-08-27	27-08-27	38														
14.1.2	A14015	Non-destructive testing — ultrasonic pulse velocity and rebound hammer	4	28-08-27	01-09-27	38														
14.1.3	A14020	Dimensional and as-built survey of the completed structure	5	10-09-27	15-09-27	26														
14.1.4	A14025	Watertightness inspection of the completed envelope	5	18-09-27	23-09-27	19														
14.2		Services testing		20-08-27	21-09-27															
14.2.1	A14030	Drainage testing — pipes, gullies, chambers, pump duty and standby	5	16-09-27	21-09-27	13														
14.2.2	A14035	Electrical testing — insulation, continuity, polarity, protection	5	02-09-27	07-09-27	33														
14.2.3	A14040	Earth resistance measurement — 5 ohm target	2	20-08-27	21-08-27	47														
14.3		Protective-system testing		05-08-27	28-09-27															
14.3.1	A14045	EMP shielding effectiveness survey — 80 dB, 10 kHz to 1 GHz	8	18-09-27	27-09-27	8														
14.3.2	A14050	Blast doors 1 and 2 — functional and seal testing	4	18-09-27	22-09-27	7														
14.3.3	A14055	Escape shaft access and closure functional test	2	05-08-27	06-08-27	60														
14.3.4	A14065	Gas-tightness and overpressure test of the envelope, bays 1–6	5	23-09-27	28-09-27	7														
14.4		Commissioning		19-08-27	06-10-27															
14.4.1	A14060	HVAC and CBRN filter train commissioning — airflow and overpressure	8	02-09-27	10-09-27	17														
14.4.2	A14070	Generator load test and changeover proving	3	19-08-27	21-08-27	39														
14.4.3	A14075	Integrated systems commissioning — protective mode operation	6	29-09-27	06-10-27	7														
14.5		Completion and handover		07-10-27	20-11-27															
14.5.1	A14080	Snagging inspection and defects list	5	18-10-27	22-10-27	0														
14.5.2	A14085	Rectification of defects	10	23-10-27	03-11-27	0														
14.5.3	A14090	As-built drawings, O&M manuals and test certificates	12	07-10-27	20-10-27	7														
14.5.4	A14095	Operator training — CBRN plant, pumps, generator, doors, escape shafts	5	21-10-27	26-10-27	7														
14.5.5	A14098	Programme contingency — festival holidays and weather allowance	10	04-11-27	15-11-27	0														
14.5.6	A14100	Final inspection and handover documentation	4	16-11-27	19-11-27	0														
14.5.7	A14105	MILESTONE: PRACTICAL COMPLETION AND HANDOVER		20-11-27	20-11-27	0														
LEGEND							■ WBS summary ■ Activity ■ Critical (TF = 0) ■ Milestone Total float													